

A Low-Power 2-bit 20MS/s Flash ADC in 45-nm CMOS with Rail-to-Rail Comparators

Bright Nnadi, Baliraja Nemade, Nithin Kidangazhiath Mana

Introduction:

Mimicking the design of the Flash ADC implemented in 65nm CMOS technology as described in [1], we developed our own version using a different PDK with a supply voltage (VDD) of 1.0V, targeting fabrication in a 45nm CMOS process. Our design employs square transistors with dimensions of $120\text{nm} \times 120\text{nm}$ and operates at a clock frequency of 20 MS/s.

While the sample-and-hold (S/H) schematic specifies a capacitor value of 300 fF, we opted to use a smaller value of 50 fF to better balance the discretization between the rising and falling edges of the input signal. Additionally, we introduced metal-oxide capacitors (MOSCAPs) to the rail-to-rail voltage divider to improve impedance matching between the voltage output and the comparator input.

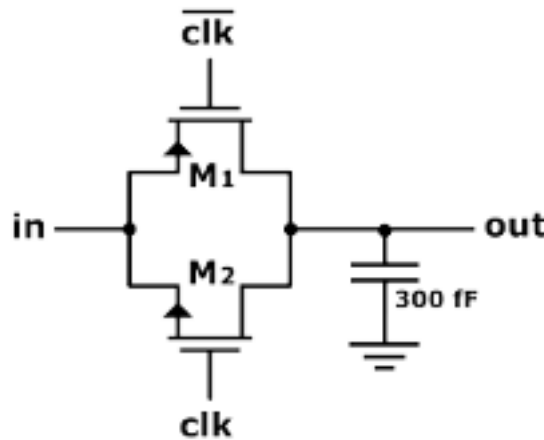


Fig 1. Schematics of Sample-hold

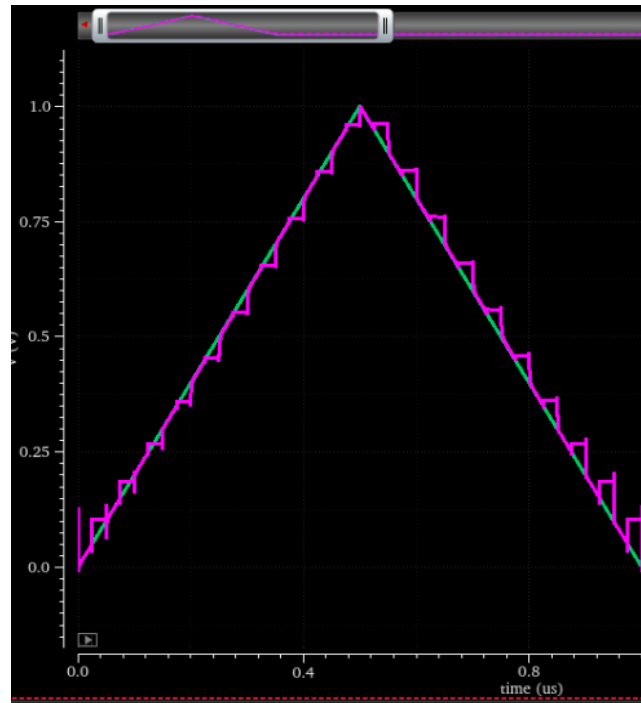


Fig 3: Input signal in green and S/H output in purple

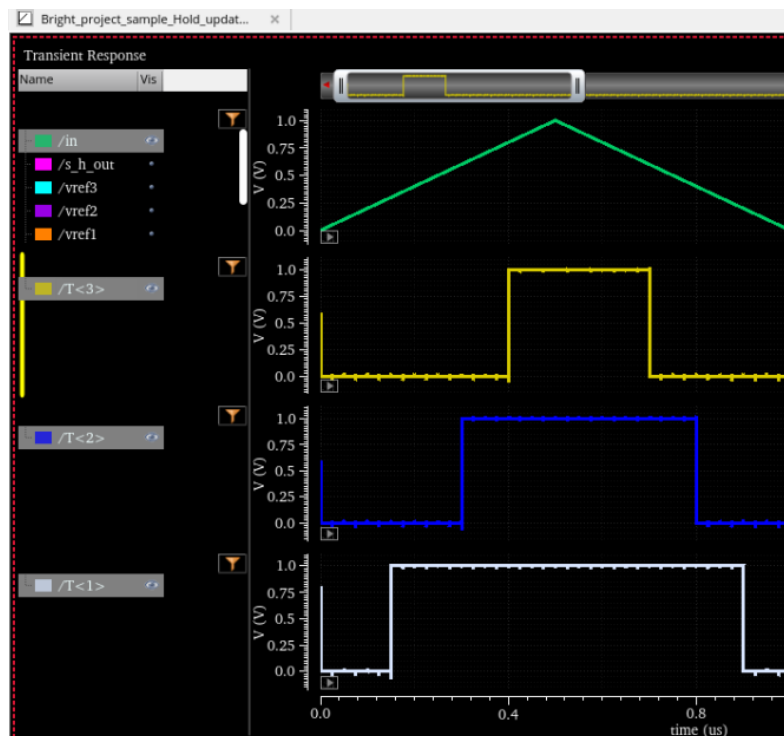


Fig 4: Input signal in green and thermometer code $\langle T3(msb):T1(lsb) \rangle$

Layout:

Layout for the full circuit is created and cleared DRC and LVS check. Following are some parts of the layout:

1. Comparator:

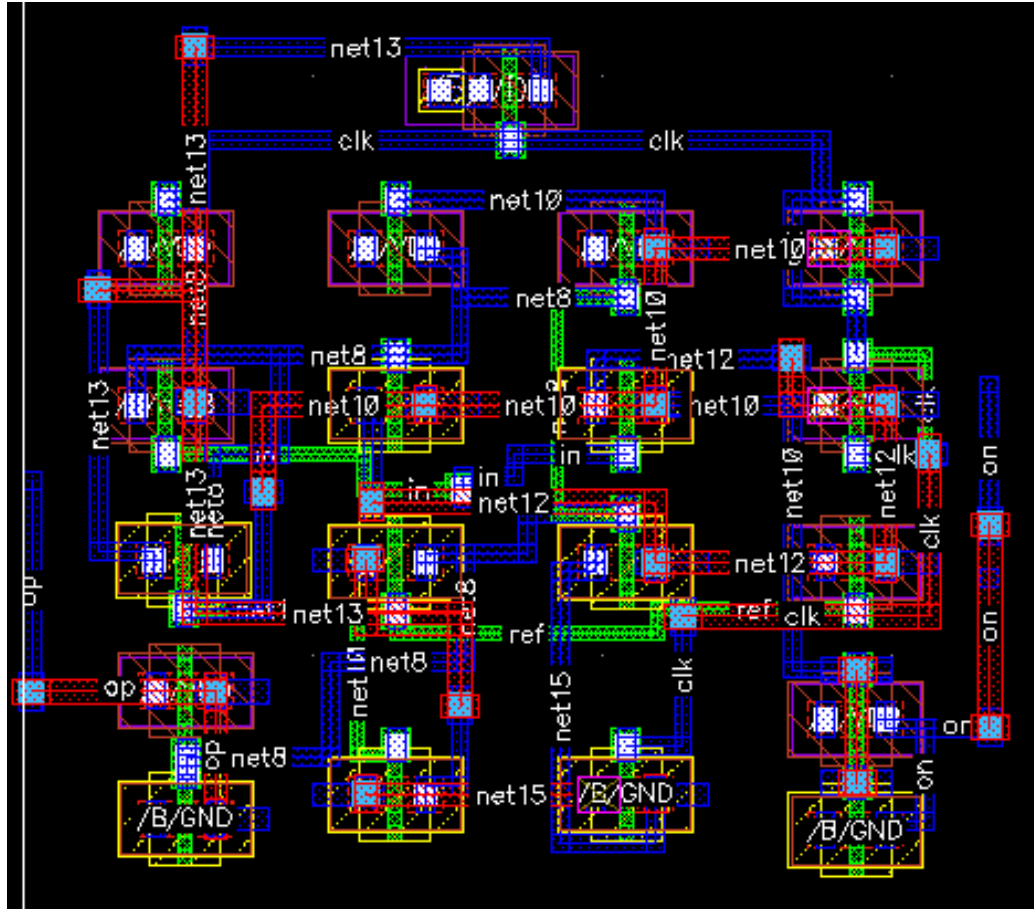


Fig 5: Comparator layout

2. SR Latch:

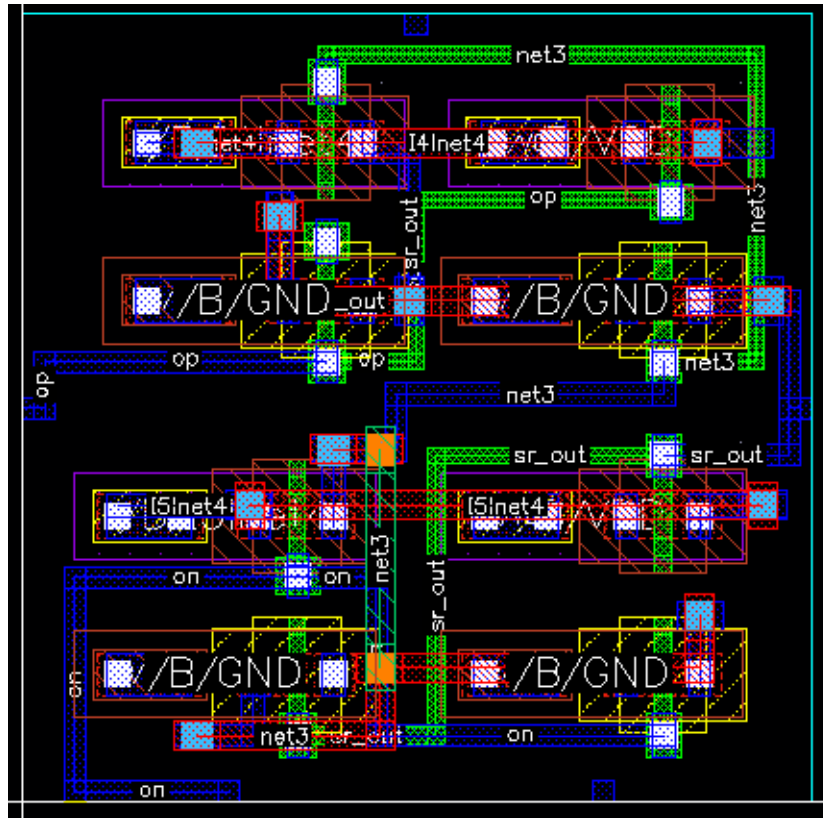


Fig 6: SR Latch layout

Post Layout Simulation:

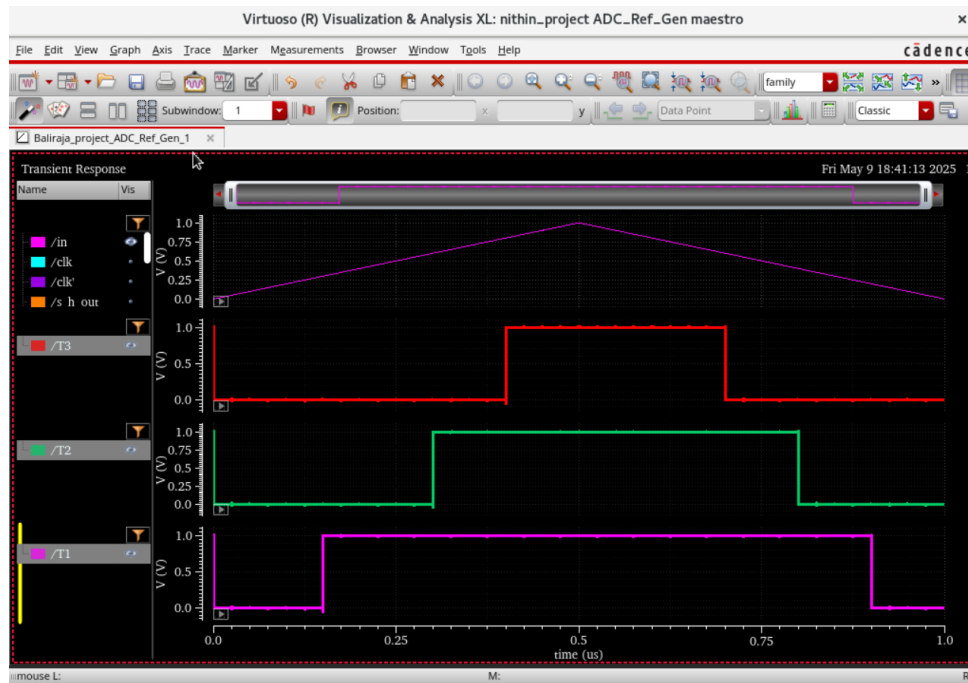


Fig 7: Input signal in green and thermometer code $\langle T3(msb):T1(lsb) \rangle$ (LayoutPS)

References:

1. T. C. Dos Santos, A. G. Girardi and P. C. C. De Aguirre, "A Low-Power 4-Bit 20-MS/s Flash ADC in 65-nm CMOS with Rail-to-Rail Comparators," *2024 Argentine Conference on Electronics (CAE)*, Bahía Blanca, Argentina, 2024, pp. 103-108, doi: 10.1109/CAE59785.2024.10487165.
2. B. Razavi, "The Flash ADC [A Circuit for All Seasons]," in *IEEE Solid-State Circuits Magazine*, vol. 9, no. 3, pp. 9-13, Summer 2017, doi: 10.1109/MSSC.2017.2712998
3. GPDK045 Reference Manual: 45nm Process Design Kit